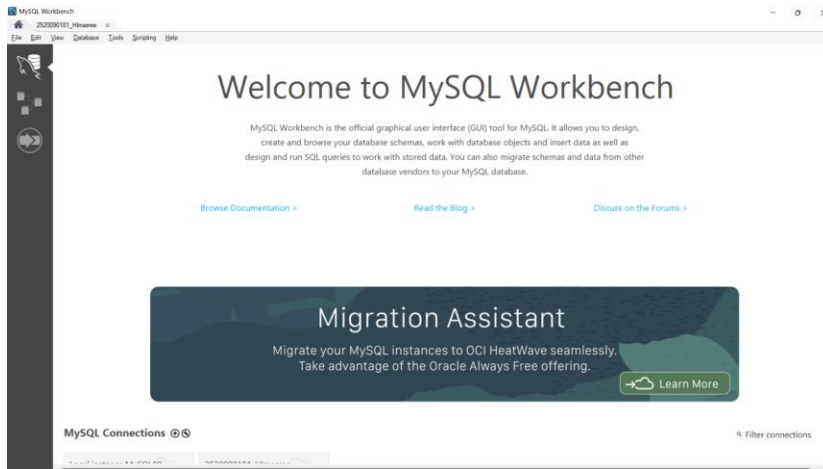
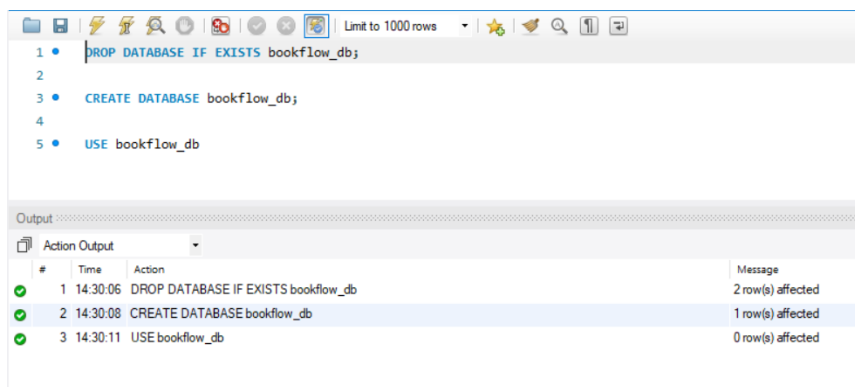


Practical-4

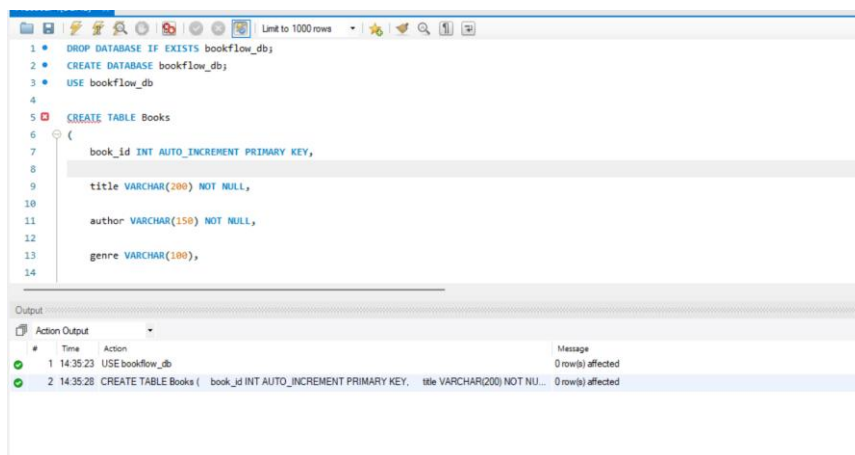
PART 1 – OPEN MYSQL WORKBENCH



PART 2 – CREATE DATABASE



PART 3–CREATE BOOKS TABLE



19 • **DESCRIBE Books;**

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NULL</u>	auto_increment
title	varchar(200)	NO		<u>NULL</u>	
author	varchar(150)	NO		<u>NULL</u>	
genre	varchar(100)	YES		<u>NULL</u>	
description	text	YES		<u>NULL</u>	

Result 1 x

Output

Action Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned

PART 4 – INSERT BOOK DATA

20 • **INSERT INTO Books**

(

title,

author,

genre,

description,

published_year

)

VALUES

(

'The Martian',

'Andy Weir',

)

Output

Action Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

PART 5 – DISPLAY BOOKS

109 • **SELECT**

book_id,

title,

author,

genre,

published_year

FROM Books

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: |

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925

Books 2 x

Output

Action Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned

PART 6 – ADD EMBEDDING COLUMN

```
1 14:36:14 ALTER TABLE Books ADD COLUMN embedding JSON
2 14:36:03 FROM Books ORDER BY book_id
3 14:36:21 SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...
4 14:36:21 INSERT INTO Books ( title, author, genre, description, published_year) VALUES ( 'The Martian'...
5 14:36:02 DESCRIBE Books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...
6 14:36:58 CREATE TABLE Books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...
7 14:36:53 USE bookflow_db
8 14:36:53 DESCRIBE Books
```

119 • DESCRIBE Books;

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			

Result 3 ×

Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned

PART 7 – INSERT MOCK VECTORS

```
120 • UPDATE Books
121 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
122 WHERE book_id = 1;
```

Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax; check the manu
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned
9	14:40:13	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```
123 • UPDATE Books
124 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
125 WHERE book_id = 2;
```

Output

#	Time	Action	Message
1	14:35:23	USE bookflow_db	0 row(s) affected
2	14:35:28	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1...	10 row(s) returned
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax; check the manual #
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned
9	14:40:13	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	14:40:34	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```

127 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
128 WHERE book_id = 3;
129 • UPDATE Books
130 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
131 WHERE book_id = 4;
132 • UPDATE Books
133 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
134 WHERE book_id = 5;
135 • UPDATE Books
136 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
137 WHERE book_id = 6;
138 • UPDATE Books
139 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
140 WHERE book_id = 7;

```

Output

#	Time	Action	Message
3	14:36:05	DESCRIBE Books	6 row(s) returned
4	14:36:51	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Mar...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0...	10 row(s) returned
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax; check the manual that cor...
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned
9	14:40:13	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	14:40:34	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
11	14:40:59	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
12	14:41:08	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	14:41:19	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	14:41:32	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
15	14:41:44	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	14:41:57	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```

142 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
143 WHERE book_id = 8;
144 • UPDATE Books
145 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
146 WHERE book_id = 9;
147 • UPDATE Books
148 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
149 WHERE book_id = 10;

```

Output

#	Time	Action	Message
5	14:37:27	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0...	10 row(s) returned
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned
9	14:40:13	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	14:40:34	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
11	14:40:59	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
12	14:41:08	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	14:41:19	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	14:41:32	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
15	14:41:44	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	14:41:57	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	14:42:28	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
18	14:42:38	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

PART 8 – VERIFY ALL VECTORS

```

150 • SELECT
151 book_id,
152 title,
153 embedding
154 FROM Books
155 ORDER BY book_id;
156

```

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]

Output

#	Time	Action	Message
6	14:38:07	FROM Books ORDER BY book_id	Error Code: 1064. You have an error in your SQL syntax; check the manual that cor...
7	14:38:14	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
8	14:39:04	DESCRIBE Books	7 row(s) returned
9	14:40:13	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	14:40:34	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
11	14:40:59	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
12	14:41:08	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
13	14:41:19	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	14:41:32	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
15	14:41:44	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	14:41:57	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	14:42:28	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
18	14:42:38	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
19	14:43:08	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned

PART 9 – SEARCH VECTOR

For this practical we use:

[0.90, 0.80, 0.20]

PART 10 – L2 DISTANCE

L2 formula:

Distance =

$$\text{SQRT}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)$$

The screenshot shows a table with 5 columns: book_id, title, author, genre, and l2_distance. The data is as follows:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0725
2	Dune	Frank Herbert	Science Fiction	0.0866

Below the table, the 'Action Output' section shows a series of SQL commands and their results:

- 8 14:39:04 DESCRIBE Books: 7 row(s) returned
- 9 14:40:13 UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 10 14:40:34 UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 11 14:40:59 UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 12 14:41:08 UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 13 14:41:19 UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 14 14:41:32 UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 15 14:41:44 UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 16 14:41:57 UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 17 14:42:28 UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 18 14:42:38 UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 19 14:43:08 SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000: 10 row(s) returned
- 20 14:46:40 SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000: 10 row(s) returned
- 21 14:46:40 SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(... 3 row(s) returned

PART 11 – COSINE SIMILARITY

Formula: $A * B / |A| * |B|$

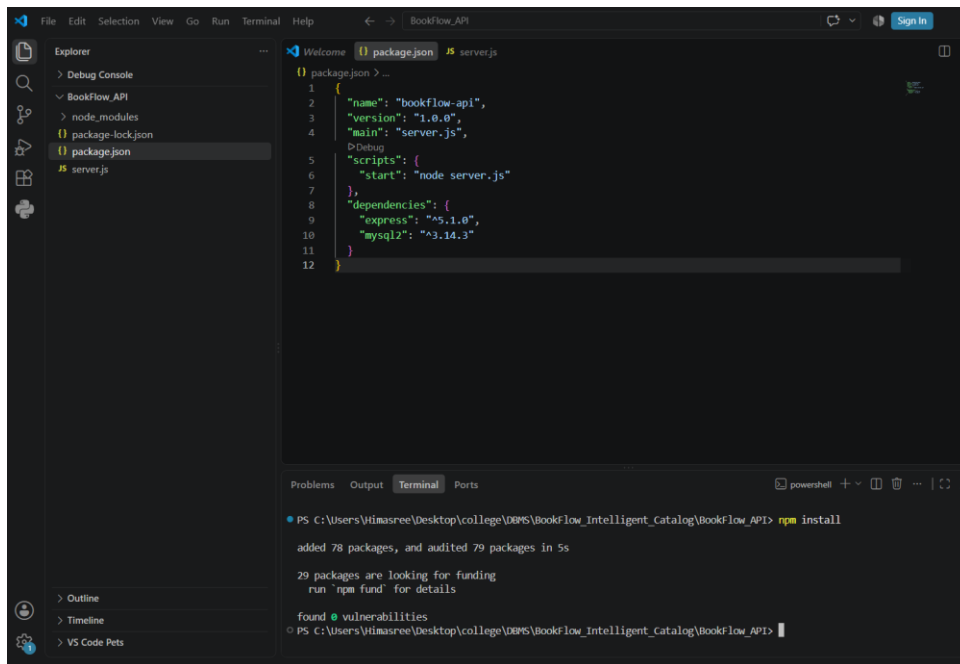
The screenshot shows a table with 5 columns: book_id, title, author, genre, and cosine_similarity. The data is as follows:

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Below the table, the 'Action Output' section shows a series of SQL commands and their results:

- 9 14:40:13 UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 10 14:40:34 UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 11 14:40:59 UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 12 14:41:08 UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 13 14:41:19 UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 14 14:41:32 UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 15 14:41:44 UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 16 14:41:57 UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 17 14:42:28 UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 18 14:42:38 UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10: 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- 19 14:43:08 SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000: 10 row(s) returned
- 20 14:46:40 SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000: 10 row(s) returned
- 21 14:46:40 SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(... 3 row(s) returned
- 22 14:47:24 SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTR... 3 row(s) returned

PART 12 – CREATE API BACKEND



```
File Edit Selection View Go Run Terminal Help
BookFlow_API

Welcome package.json JS server.js

1 {
2   "name": "bookflow-api",
3   "version": "1.0.0",
4   "main": "server.js",
5   "scripts": {
6     "start": "node server.js"
7   },
8   "dependencies": {
9     "express": "^5.1.0",
10    "mysql2": "^3.14.3"
11  }
12 }
```

Problems Output Terminal Ports

PS C:\Users\Himasree\Desktop\college\DBMS\BookFlow_Intelligent_Catalog\BookFlow_API> npm install

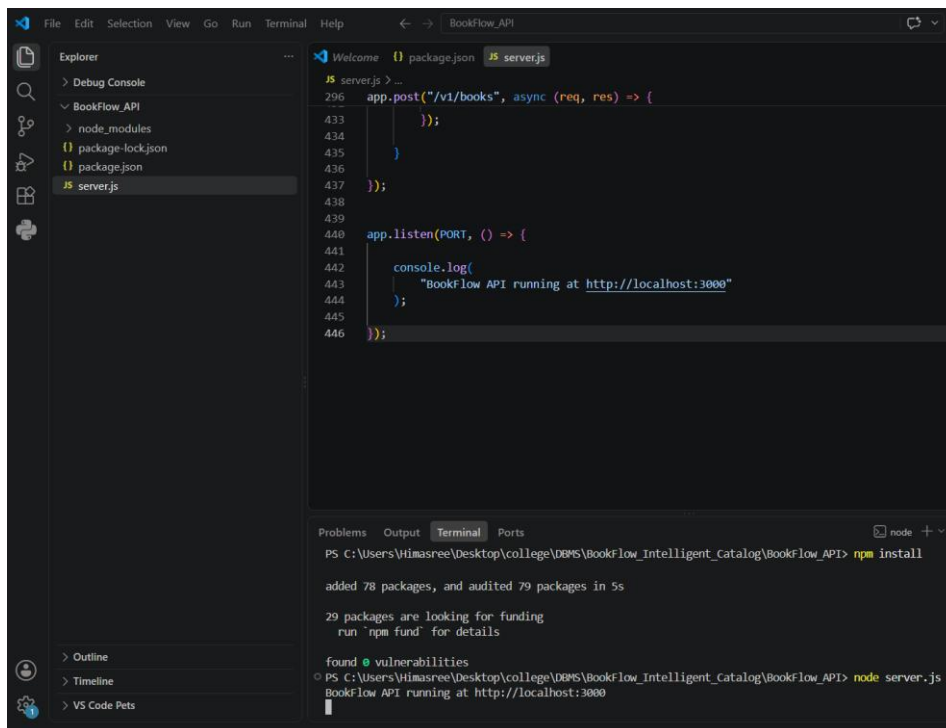
added 78 packages, and audited 79 packages in 5s

29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities

PS C:\Users\Himasree\Desktop\college\DBMS\BookFlow_Intelligent_Catalog\BookFlow_API>

PART 13 – START NODE.JS SERVER



```
File Edit Selection View Go Run Terminal Help
BookFlow_API

Welcome package.json JS server.js

296 app.post("/v1/books", async (req, res) => {
433   });
434 }
435 }
436 });
437 });
438
439 app.listen(PORT, () => {
440   console.log(
441     "BookFlow API running at http://localhost:3000"
442   );
443 });
444
445
446 });
```

Problems Output Terminal Ports

PS C:\Users\Himasree\Desktop\college\DBMS\BookFlow_Intelligent_Catalog\BookFlow_API> npm install

added 78 packages, and audited 79 packages in 5s

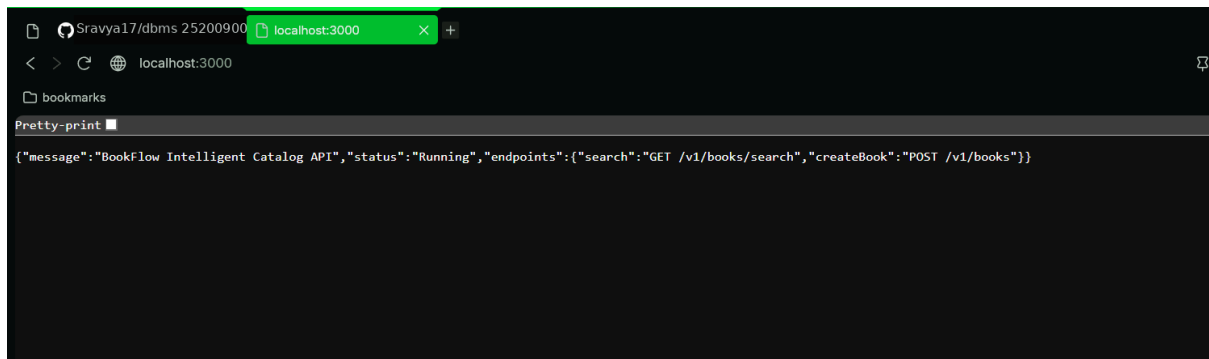
29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities

PS C:\Users\Himasree\Desktop\college\DBMS\BookFlow_Intelligent_Catalog\BookFlow_API> node server.js

BookFlow API running at http://localhost:3000

PART 14 – TEST API IN BROWSER



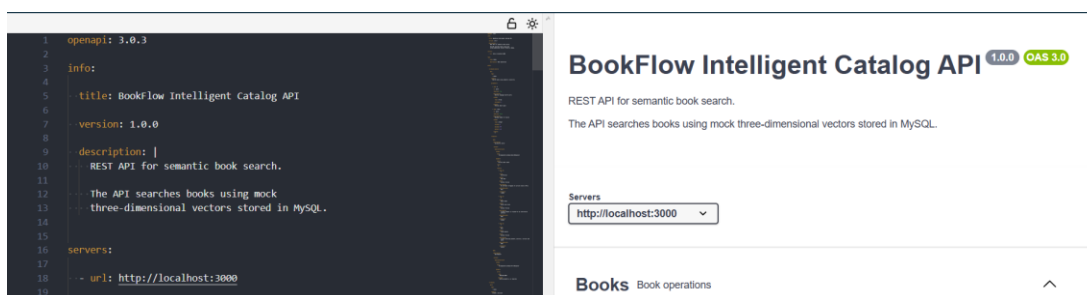
PART 15 – SWAGGER EDITOR

```

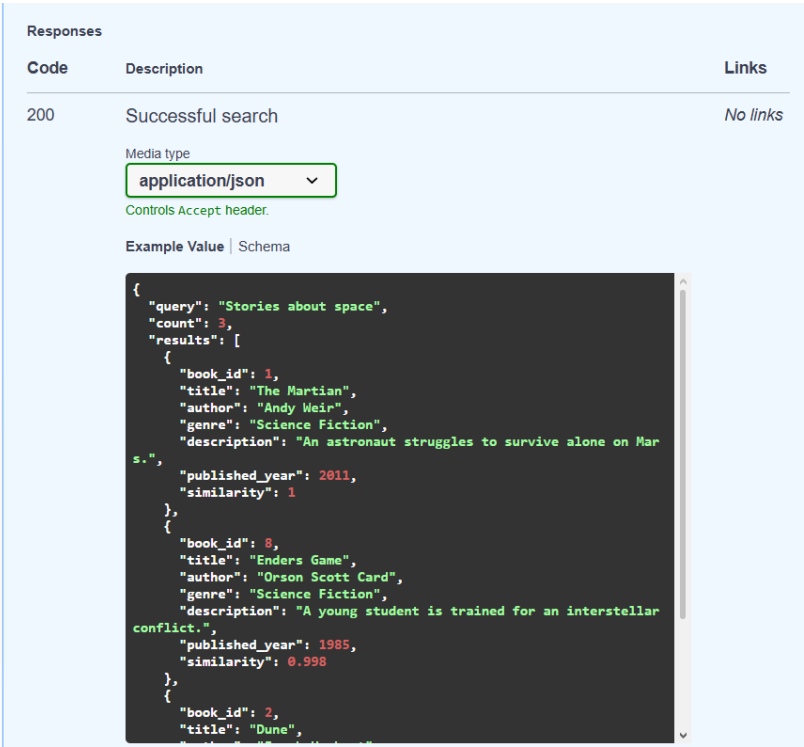
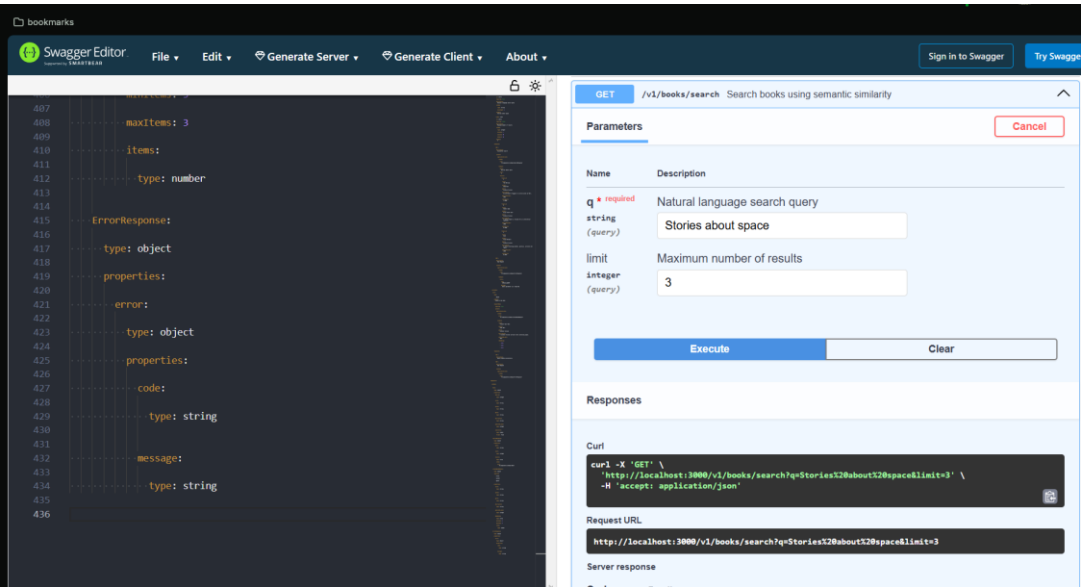
1  openapi: 3.0.3
2
3  info:
4
5    title: BookFlow Intelligent Catalog API
6
7    version: 1.0.0
8
9    description: |
10     REST API for semantic book search.
11
12     The API searches books using mock
13     three-dimensional vectors stored in MySQL.
14
15
16  servers:
17
18    url: http://localhost:3000
19
20
21  tags:
22
23    name: Books
24
25    description: Book operations
26
27
28  paths:
29
30
31    /v1/books/search:
32
33      get:
34
35        tags:

```

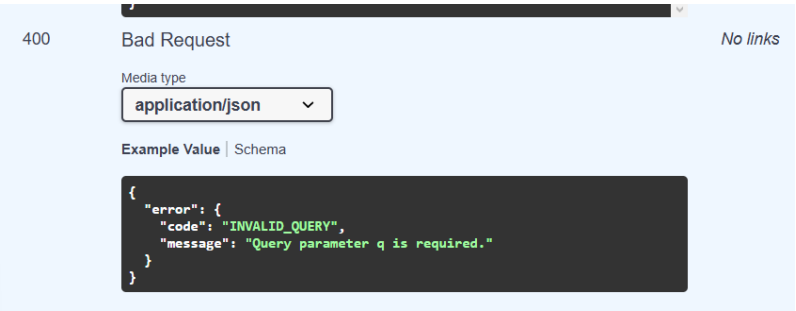
PART 16 – SWAGGER EDITOR OUTPUT



PART 17 – SWAGGER TEST GET



PART 18 – SWAGGER TEST 400 ERROR



PART 19 – SWAGGER TEST POST

Parameters

Cancel

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Project Hail Mary",
  "author": "Andy Weir",
  "genre": "Science Fiction",
  "description": "A science fiction survival story involving space.",
  "published_year": 2021,
  "embedding": [
    0.93,
    0.82,
    0.12
  ]
}
```

Execute

Clear

Curl

```
curl -X 'POST' \
  'http://localhost:3000/v1/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Project Hail Mary",
    "author": "Andy Weir",
    "genre": "Science Fiction",
    "description": "A science fiction survival story involving space.",
    "published_year": 2021,
    "embedding": [
      0.93,
      0.82,
      0.12
    ]
  }'
```

Request URL

http://localhost:3000/v1/books

Server response

Code

Details

Undocumented

Failed to fetch.

Possible Reasons:

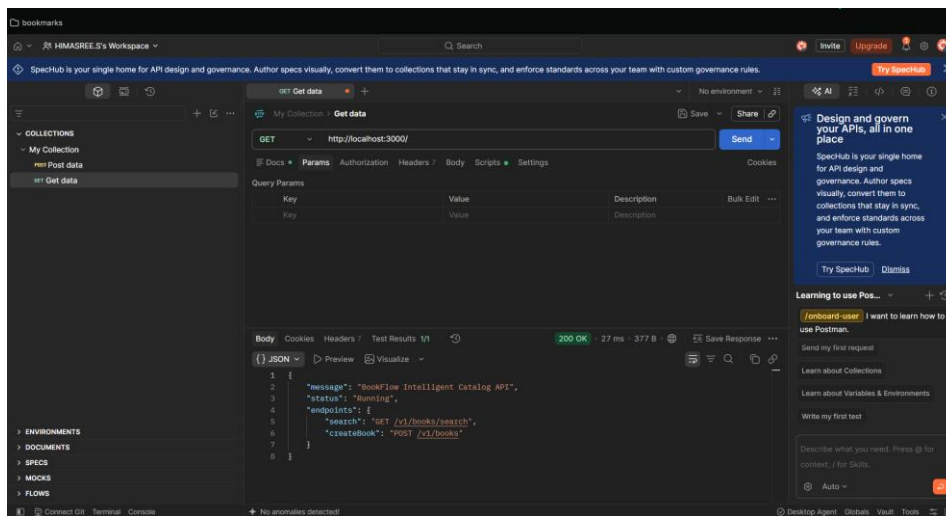
- CORS
- Network Failure
- URL scheme must be "http" or "https" for CORS request.

Responses

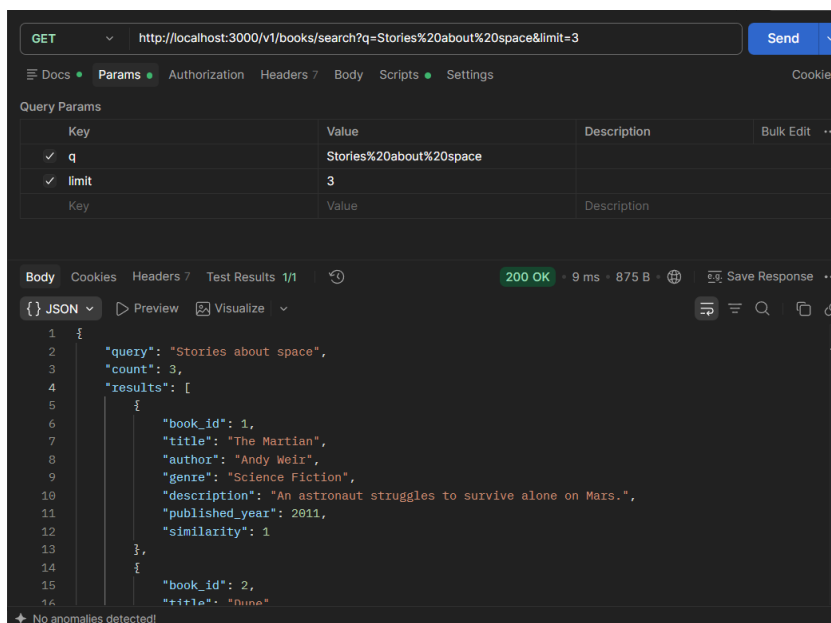
Code	Description	Links
201	Book created successfully	No links

PART 20 – POSTMAN TESTING

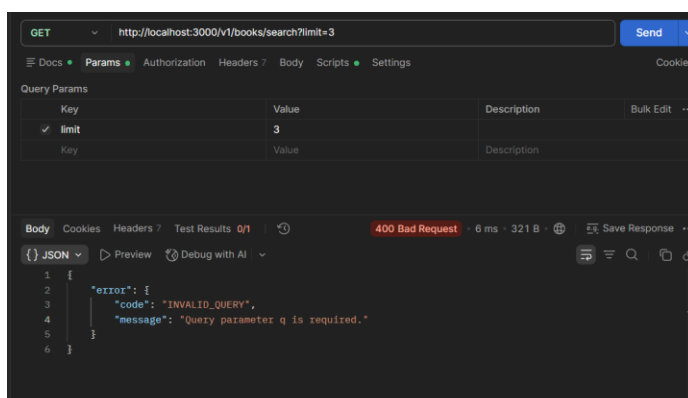
TEST 1 – API STATUS



TEST 2 – SEMANTIC SEARCH



TEST 3 – MISSING q



TEST 4 – INVALID LIMIT

GET `http://localhost:3000/v1/books/search?q=space&limit=100`

Query Params

Key	Value	Description
q	space	
limit	100	

Body

400 Bad Request • 4 ms • 322 B

```
{
  "error": {
    "code": "INVALID_LIMIT",
    "message": "limit must be between 1 and 50."
  }
}
```

TEST 5 – CREATE BOOK

GET `http://localhost:3000/v1/books`

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body

404 Not Found • 4 ms • 419 B

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Error</title>
</head>
<body>
<pre>Cannot GET /v1/books</pre>
</body>
</html>
```

TEST 6 – INVALID POST

GET `http://localhost:3000/v1/books`

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body

404 Not Found • 3 ms • 419 B

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Error</title>
</head>
<body>
<pre>Cannot GET /v1/books</pre>
</body>
</html>
```

PART 21 – VERIFY DATA AFTER POST

311 • SELECT
312 book_id,
313 title,
314 author.

book_id	title	author	genre	published_year	embedding
1	The Martian	Andy Weir	Science Fiction	2011	[0.90, 0.80, 0.20]
2	Dune	Frank Herbert	Science Fiction	1965	[0.85, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	Douglas Adams	Science Fiction	1979	[0.80, 0.70, 0.30]
4	Pride and Prejudice	Jane Austen	Romance	1813	[0.10, 0.20, 0.90]
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925	[0.15, 0.30, 0.80]

Books 8 x

Output

#	Time	Action	Message
✓	10 14:40:34	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	11 14:40:59	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	12 14:41:08	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	13 14:41:19	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.80, 0.20) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	14 14:41:32	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	15 14:41:44	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	16 14:41:57	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	17 14:42:28	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	18 14:42:38	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	19 14:43:08	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
✓	20 14:46:40	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
✓	21 14:46:40	SELECT book_id, title, author, genre, ROUND(SORT(POW(CAST(	3 row(s) returned
✓	22 14:47:24	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTR	3 row(s) returned
✓	23 15:30:05	SELECT book_id, title, author, genre, published_year, embedding FROM Books ORDER BY bo	10 row(s) returned